

Segmenting OT Assets That Cannot Run Agents

The most common objection to network microsegmentation in manufacturing: *'Our PLCs, CNC machines, and industrial robots run proprietary firmware. We cannot install agents. We cannot disrupt production.'*

Akamai Guardicore Segmentation, now with NVIDIA BlueField DPU integration (GA Q2 2026), eliminates this objection entirely.

The Problem: Un-Agentable OT Assets in Every Production Floor

- Siemens SIMATIC PLCs and S7 controllers
- KUKA industrial robots and their controllers
- Fanuc / Mitsubishi CNC machine controllers
- HMI panels (Windows CE / embedded Linux, vendor-locked)
- Legacy SCADA systems on Windows XP / 2003
- Embedded sensors and IoT gateways
- Automated Guided Vehicles (AGVs)

The Akamai + NVIDIA BlueField Solution (GA Q2 2026)

The NVIDIA BlueField DPU (Data Processing Unit) is a SmartNIC that handles network policy enforcement at the hardware layer — below the operating system, without any software on the end device.

How it works	Benefit
BlueField DPU installed on server adjacent to OT network switch	Hardware layer — no OT device touch
Guardicore agent on the BlueField DPU (not on OT devices)	All enforcement in the SmartNIC
Guardicore discovers all OT flows via passive network observation	Complete visibility of all machines
Segmentation policy applied without touching any production device	Zero downtime; zero vendor approval needed
All logs and flow data reported to central Guardicore console	Unified IT + OT visibility dashboard

Why This Matters for NIS2 and TISAX

NIS2 Article 21 and TISAX 6.0 IS-19 both require documented segmentation of IT/OT boundaries — without exception for legacy equipment. The BlueField solution enables manufacturers to satisfy these requirements on 100% of their production floor, including assets that have been exempt from previous segmentation projects due to agent constraints.

Contact Axians IT Security Services for a BlueField-enabled OT segmentation assessment: info@axians.de